A GORENSTEIN MULTICROSS SINGULARITY OVER AN ALGEBRAICALLY CLOSED FIELD IS A NODE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file curves.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the completion reduction, the n = 2 complete-intersection case, the full Ext^1 computation with the coordinate functionals e_i, the independence check, and the final dimension contradiction are all retained verbatim.

Editorial note. The label lemma-multicross-gorenstein-is-nodal names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The multicross normal form. Same chapter, Definition definition-multicross and Equation equation-multicross. Provides the completed local normal form A with n branches whose Gorenstein property is analysed.
- Gorenstein permanence under completion. More on Algebra, Section more-algebra-section-permanence-completion; Dualizing Complexes, Lemma dualizing-lemma-flat-under-gorenstein. The completed local map is flat and unramified with trivial residue extension, transferring the Gorenstein property from O_{X,x} to its completion.
- Duality inputs. Duality for Schemes, Definition duality-definition-gorenstein, Lemma duality-lemma-gorenstein-lci, Lemma duality-lemma-find-function; Dualizing Complexes, Lemma dualizing-lemma-regular-gorenstein. Handles the n = 2 case (complete intersection) and, for n > 2, the constraint R Hom(k, A) = k[n] on a Gorenstein ring.

Source reference. Original reference 0C1W: sources/stacks-curves/pinned-original-source.tex, source lines 3463–3463. Prerequisite pointer invoked at proof line(s) 3538; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original definition 0AWW: sources/stacks-chow/tags.txt, source lines 13734–13734. Prerequisite pointer invoked at proof line(s) 3565; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BVA: sources/stacks-chow/tags.txt, source lines 14903–14903. Prerequisite pointer invoked at proof line(s) 3557; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

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Source reference. Original lemma 0A7L: sources/stacks-chow/tags.txt, source lines 12884–12884. Prerequisite pointer invoked at proof line(s) 3567; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BJL: sources/stacks-chow/tags.txt, source lines 14529–14529. Prerequisite pointer invoked at proof line(s) 3549; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AWX: sources/stacks-chow/tags.txt, source lines 13735–13735. Prerequisite pointer invoked at proof line(s) 3560; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original equation 0C1U: sources/stacks-curves/pinned-original-source.tex, source lines 3410–3410. Prerequisite pointer invoked at proof line(s) 3551; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 07NU: sources/stacks-chow/tags.txt, source lines 9774–9774. Prerequisite pointer invoked at proof line(s) 3546; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let k be an algebraically closed field. Let X be a reduced algebraic 1-dimensional k-scheme. Let $x \in X$ be a multicross singularity (Definition 0C1W). If X is Gorenstein, then x is a node.*

Proof. The map $\mathcal{O}_{X,x} \to \mathcal{O}^\wedge_{X,x}$ is flat and unramified in the sense that $\kappa(x) = \mathcal{O}^\wedge_{X,x}/\mathfrak{m}_x\mathcal{O}^\wedge_{X,x}$. (See More on Algebra, Section 07NU.) Thus X is Gorenstein implies $\mathcal{O}_{X,x}$ is Gorenstein, implies $\mathcal{O}^\wedge_{X,x}$ is Gorenstein by Dualizing Complexes, Lemma 0BJL. Thus it suffices to show that the ring A in (0C1U) with $n \geq 2$ is Gorenstein if and only if $n = 2$.

If $n = 2$, then $A = k[[x, y]]/(xy)$ is a complete intersection and hence Gorenstein. For example this follows from Duality for Schemes, Lemma 0BVA applied to $k[[x, y]] \to A$ and the fact that the regular local ring $k[[x, y]]$ is Gorenstein by Dualizing Complexes, Lemma 0AWX.

Assume $n > 2$. If A where Gorenstein, then A would be a dualizing complex over A (Duality for Schemes, Definition 0AWW). Then $R\operatorname{Hom}(k, A)$ would be equal to $k[n]$ for some $n \in \mathbf{Z}$, see Dualizing Complexes, Lemma 0A7L. It would follow that $\operatorname{Ext}^1_A(k, A) \cong k$ or $\operatorname{Ext}^1_A(k, A) = 0$ (depending on the value of n; in fact n has to be -1 but it doesn't matter to us here). Using the exact sequence

$$0 \to \mathfrak{m}_A \to A \to k \to 0$$

we find that

$$\operatorname{Ext}^1_A(k, A) = \operatorname{Hom}_A(\mathfrak{m}_A, A)/A$$

where $A \to \operatorname{Hom}_A(\mathfrak{m}_A, A)$ is given by $a \mapsto (a' \mapsto aa')$. Let $e_i \in \operatorname{Hom}_A(\mathfrak{m}_A, A)$ be the element that sends $(f_1, \ldots, f_n) \in \mathfrak{m}_A$ to $(0, \ldots, 0, f_i, 0, \ldots, 0)$. The reader verifies easily that $e_1, \ldots, e_{n-1}$ are k-linearly independent in $\operatorname{Hom}_A(\mathfrak{m}_A, A)/A$. Thus $\dim_k \operatorname{Ext}^1_A(k, A) \geq n - 1 \geq 2$ which finishes the proof. (Observe that $e_1 + \ldots + e_n$ is the image of 1 under the map $A \to \operatorname{Hom}_A(\mathfrak{m}_A, A)$.) $\square$